Z. Tierpsychol., **30**, 277—296, (1972)
© 1972 Verlag Paul Parey, Berlin und Hamburg

Zoology Department, Duke University, Durham, North Carolina, U.S.A.

Patterns of Maternal Care in Lemurs: II. Effects of Group Size and Early Separation

By PETER H. KLOPFER

With 12 figures

Received: 10. 3. 1971

Introduction

"The lability of social structures across a species range, together with their relative local stability, suggests that they are controlled and maintained partly through direct sociological response to environment and partly through social traditions. Primate societies are thus conceived as 'protocultural' communities subject to processes of historical change additional to Darwinian selection of inherited traits. Social conformity and ecological adaptation are viewed within this context as being largely 'shaped' by the environment. The habitat, physical and social are, as it were, nature's Skinner box, providing locally characteristic types and frequencies of natural learning trials in complex schedules that inter-relate to pattern individual behaviour as a whole." (CROOK 1970, p. 130).

The conception of the interplay between ecological and behavioral features of primate societies suggested by this excerpt raises questions of especial interest when the prosimian primates are considered. The Lemuriformes are almost certainly phyletically closer to one another than are, *inter se*, the species of monkeys or apes that generally serve for laboratory studies (BUETTNER-JANUSCH 1966 and MS). Yet, the diversity of prosimian habitats and life styles is astounding. Considering but one aspect of their social deportment, *Daubentonia* and *Lepilemur* have only been reported as solitary animals (PETTER 1965). *Microcebus* occasionally forms sleeping groups, *Propithecus* and *Indri* form family groups, while species of the genus *Lemur* form even larger societies (*ibid*). This does not by any means exhaust the list of differences. The list suffices to suggest why these Madagascar primates provide such intriguing material for those of us interested in the interface of behavior and ecology.

The purposes of our studies, of which this is the second report (for number one, see KLOPFER & KLOPFER 1970), are to uncover the factors that determine whether a mother will care for her young and whether that care is restricted exclusively to her own or extends to alien youngsters. From the standpoint of the infant, we are interested in knowing how a separation from the mother affects its development under circumstances where a surrogate mother may or may not be available. Finally, we want to examine the long-term results of different rearing conditions. All of this is to be studied against the different social backgrounds provided by three particular species of *Lemur:*

catta, fulvus, and *variegatus*. These three, while similar in their morphology, and, we assume, their physiology, are neither truly sympatric nor similar in their patterns of maternal-care (*ibid*). In *catta,* animals other than the mother may contact the infant before it is a week old and independent exploration begins before it has aged 3 weeks. In *fulvus,* however, except for the mate, other animals are generally prevented by the mother from taking over the infant until 4 weeks after its birth, simultaneously with the commencement of independent explorations by the infant. The *variegatus* infant, finally, is denied early contact of any kind, except with the mother, until it is fully mobile, at about 7—8 weeks. Before this, it receives only intermittent attention even from its mother.

What is the significance of these different patterns of care? Are the more communally reared *catta* infants better buffered from the trauma of parental separation? Are the "independent" *variegatus* better able to invade and colonize new areas? These queries represent the direction in which these studies will lead.

Natural observations on Mother-Infant Contacts

To what extent do the conditions of their captivity distort the behavior of our lemurs?

Foremost among the difficulties of dealing with captive, lab-reared animals are the distortions of the usual patterns of behavior induced by confinement. The design of the Primate Facility was expected to minimize such effects, in part by providing considerable indoor space, and in part by allowing access to large outdoor runs (Fig. 1). The indoor space consists of hexagonal rooms, 5.3 m. across (as measured between opposite corners) and almost 5 m. high (Fig. 2); the adjoining runs cover an area of about 100 and 150 m^2, respectively. The hexagonal rooms, aside from their providing considerably more space than the usual cages employed for lemurs (about 1 $^1/_2$ m^3) lack acute angle. The inner space is virtually cornerless, which probably makes it functionally more similar to an unconfined space than would a mere increase in size without elimination of corners. The general issue of the relation between space and animal spacing is treated below.

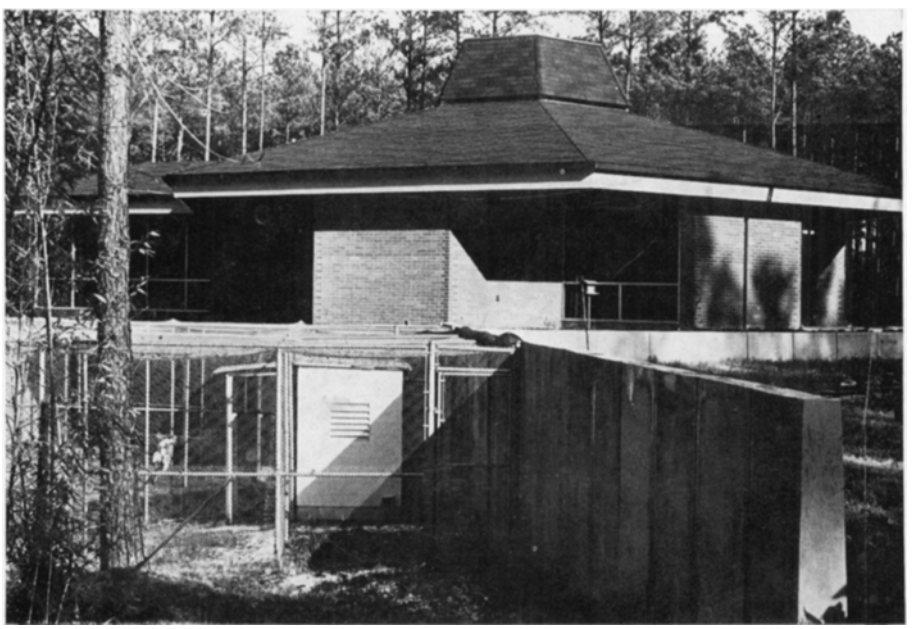


Fig. 1: View of Outdoor Run

It is apparent that comparisons with wild lemurs in their usual habitat are an essential prerequisite to the interpretation of the laboratory results. To this end, observations were made of several troops of *Lemur catta*, at Berenty, on the island Republic of Madagascar. The preserve within which the animals were studied has been described in detail by JOLLY (1966). JOLLY (*ibid*) has also described the behavior and ecology of the lemurs themselves (*ibid*). The preserve is in a climax, probably virgin, Tamarind forest, with a dense canopy. The trees rise to a height of 30 meters from the banks of the Mandrary River in the south of Madagascar. A study of the stability of the territories of the *Lemur catta* troops living there was reported by KLOPFER and JOLLY (1970). The data reported here are derived from the same area and animals. They were collected during January of 1969 by myself, with the field assistance of J. HARRINGTON (Duke University), in August—September 1969 by N. BUDNITZ (Duke University), and again in October—November 1970 by Drs. A. SHKOLNIK and U. ORGAD (University of Tel Aviv).

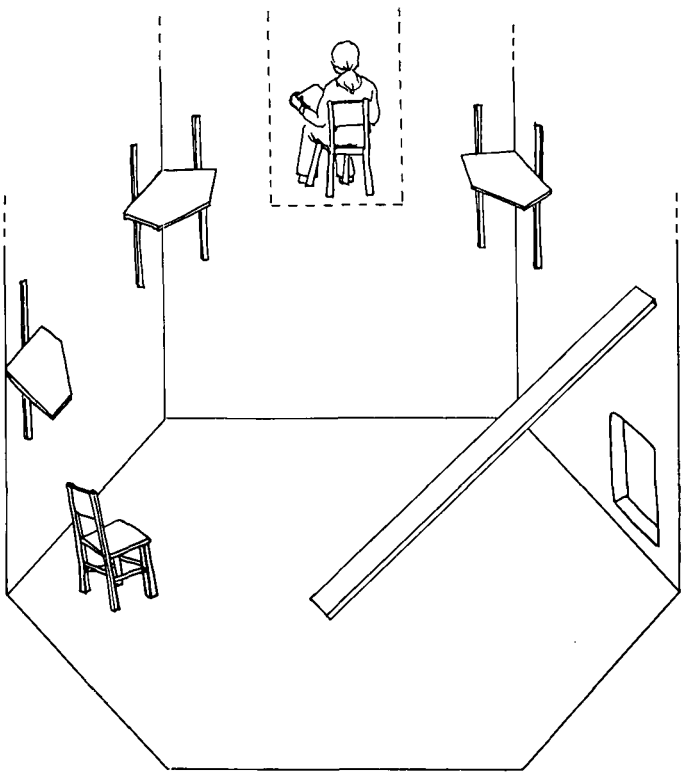


Fig. 2: Schematic of Chamber

Methods

The period of observations extended from 18 — 30 January, 1969, 26 August — 24 September, 1969, and 29 September — 5 October, 1970. A minimum of one hundred hours of contact with the lemurs was logged during each of these 3 periods. The procedure was to stalk the animals from a distance that was sufficient to avoid elicitation of the "click-grunt" alarm (see JOLLY 1966). This distance varied from 3 to over 30 meters, depending on cover. In fact, the animals take remarkably little cognizance of man, having been rarely frightened or fed by him. Adults and older juveniles frequently approached within patting distance when one sat quietly (the impulse to pet was successfully subdued). Binoculars (8×30 mm or 7×50 mm) were used for identification; distances were either paced or estimated, the estimates being periodically checked with an optical range finder. Observations began at about 500 hours, which was some 30 minutes before sunrise and extended to no later than 1800 hours (often only to 1700). The lemurs actually continued to feed until dark (1900—1930, depending on cloud cover), but rarely moved any distance after 1700. Other details are given by JOLLY (1966).

Upon locating a troop and taking up an acceptable vantage point, the observer selected the female with young nearest him, and proceeded to observe her for no less than 30 minutes. Particular animals could be recognized due to individual differences in appearance. If the pair were lost sight of prior to the completion of the 30 minute period, those observations were not included in the data summary. After any one ♀ had been observed for a total of 10 hours (spread over several days), no further data from her were included. Thus, each of the 100 hours of observation of mother-infant pairs made by September 1969 and again in September — October 1970 were distributed among at least 10 different ♀♀. No ♀ was represented by less than 30 minutes of continuous observation, nor by more than 10 hours of total observation. This procedure was developed to minimize bias that could result from our small and nonrandomly selected sample. At 30 second intervals during a 30 minute observation session observers recorded the following items: Was the infant asleep or awake (i. e., infant nursing or stirring, eyes open)? Was it in contact with its mother, some other animal, or was it separated? If separated, by what distance and for how long? Other details were recorded as opportunity provided, e. g., favored positions of the infants, activities of their mothers and their troops, and intra-troop synchrony in behavior. However, because these latter items were not systematically collected, no specific use is made of them here.

The 30 second intervals at which maternal-filial relations were sampled were indicated by a small battery-operated audio-signal marker, pocket size, with an ear-plug head set. The quality of the 30 second time signal differed from that at 60 second intervals, allowing for convenient and foolproof timing of events. (This unit was designed and constructed by J. BIRCH, Electronics Shop, Psychology Department, Duke University.)

Results

The data are summarized in Table 1.

Table 1: Mother-Infant Relations

	Number of sampling intervals with						
	infants asleep			infants awake			
	altogether separate	contacting mother	contacting another	altogether separate	contacting mother	contacting another	total
Aug. - Sept. 1969	0	896	0	2	10,622	14	11,534
Sept. - Oct. 1970	5	950	7	374	10,476	185	11,997

The first birth recorded in 1969 was on 26 August. The observer (N. BUDNITZ) had been on the scene for almost four weeks by this date, so this date probably marks the onset of the birthing season. Thereafter, births occurred in quick succession. In 1963, JOLLY recorded the first birth as occurring 20 or 21 August, then 5 others between 20—27 August. Thus, the infants observed in August — September probably ranged from 1—30 days of age, while those in the second sample, 30—40 days.

During their first month, the infants were never seen to sleep away from their mothers; only after 30 days, and then rarely, were infants detected sleeping alone or with an animal other than their own mothers. This was generally an older ♀, or another ♀ with young. The change between the first 30 days and subsequently is greatest when one examines the behavior of the wakeful infant. The frequency of contacts with other animals now rises sharply (from 14 to 185), and the periods of separation even more so, 2 to 374. It must be emphasized that the category "contacting another" includes only those episodes where there was no simultaneous contact with the mother. JOLLY (1966), too, records relatively frequent instances of mothers reciprocally grooming one another's infants, through always with the infants retaining maternal contact.

A comparison with the data from the captive lemurs in the Duke colony is instructive (and see KLOPFER & KLOPFER 1970). The start of a consistent increase in the proportion of time the infant spent off the mother (when in

captivity) occurred at about 30 days of age. Specifically, the percentage of the observation time when either of two infants was not in contact with its mother rose from less than 10% by day 28, to over 15% by day 32 and over 30% by day 48 (one of the two reached the 30% mark on day 40). Wakeful contacts with other adults, simultaneously with maternal contact, occurred from the start. Contacts with others when separated from the mother were not noted until day 8, and then only infrequently. This corresponds closely with the situation at Berenty, where a significant number of wakeful contacts with animals other than the mother were also noted only during the September—October period. The first occasion on which a captive infant moved from the ventral surface to the back of its mother was day 3; this corresponds to the earliest age noted both by JOLLY (1966) and by us at Berenty. In the colony, the first independent exploration was noted as early as day 18, though this did not occur regularly until about day 30 (day 27 and 37, respectively). At Berenty, too, only 2 occasions of total separation were noted before day 30 (approximately), and several hundred thereafter. Thus, with respect to the specific indices of development which we used, the captive and wild animals appeared strikingly similar. While subjective reports must be treated skeptically, let me nonetheless record that the observers did indeed agree that this was the case, even with regard to other, less carefully measured attributes.

Short-term Effects of Mother-Infant Separation

In a previous report (KLOPFER & KLOPFER 1970), we described the behavior of captive lemurs that were minimally disturbed during observation periods that extended for 215 days after an infant's birth. Section I of this report gives some assurance that the conditions of captivity had not unduly affected these aspects of the animals' behavior which we were examining. Obviously, we must assume there were some changes or we would have to repudiate the notion of lability in primate social behavior, the belief that ecological factors (including the relatively bizarre conditions consequent upon captivity) are reflected in behavior. However, our laboratory time scale is highly compressed as compared with that against which evolutionary changes are measured. Thus, so long as the lability of the features we study is more a function of selective than ontogenetic forces, the inevitable pathologies of captivity need not be our undoing. Where the facets we examine are so labile as to respond rapidly to the removal from the natural environment, a different research strategy is obviously required.

These studies, in any event, did proceed from the assumption that the patterns of maternal care we observed were "normal". The design called for comparisons between the 3 species of lemur (*catta, fulvus, variegatus*) under 3 sets of conditions: infant alone with mother; with mother and father; with parents and other adults and juveniles. Only *Lemur fulvus* is dealt with here. Even so, the full design could not be completed, due to deaths or reproductive failures. Table 2 indicates the identities of the animals used for this report. Data concerning the other species are not yet available.

Table 2: Design and Identities of *Lemur fulvus* Study

mother / infant	mother , father / infant	parents , group / infant
Sal[1] and Ursula[4] (♀, 1969)	Atalanta[3] , Sforza[3] and Aphrodite (♀, 1969)	Lisa[2] , Pop[1] and Corelli (♂, 1969) , with Rufina[4] (♀),
Calypso[2] and Creon[5] (♀, 1970)	Lisa , Pop and Viola (♀, 1970)	Clarissa[4] (♀) and Nausicaa[4] (♀) (in outdoor run)

[1] Considered to be *Lemur fulvus fulvus*
[2] Considered to be *Lemur fulvus rufus*
[3] Considered to be *Lemur macaco*
[4] Considered to be *Lemur fulvus fulvus - rufus* "hybrids"
[5] Considered to be *Lemur fulvus fulvus - collaris* "hybrids"

Methods

The observation chambers of the Primate Facility within which the animals lived were bare, hexagonal rooms with a volume of about 82 cubic meters. The ceiling tapered to a height of 5 meters; the maximum distance from corner to corner across the floor was 5.3 meters. Corner shelves and a single central post with several horizontal branches provided perches. Observations were made through one-way glass either from an overhead mezzanine or at ground level (see KLOPFER & KLOPFER 1970 and Fig. II). The rooms are not fully soundproof, but only especially loud sounds distract the animals. One group of 6 animals (Lisa, Pop, Corelli and 3 ♀♀) was observed while in an adjoining outdoor run, about 150 square meters in area. Here, the observer watched from inside the building, but was not visually concealed. However, these animals were thoroughly habituated to humans as consequence of an endless daily procession past their area.

The animals were maintained on a fixed schedule of 12 hours of light 12 of dark (dim red light). While nocturnal observations were made to establish activity patterns, the maternal data were collected during daylight hours. Observations were evenly spaced throughout those periods of the day when the animals were most likely be active. (Nocturnal activities were observed with a light amplification device). These observations were supplemented by videotape records, which served to confirm that periods of presumed inactivity were indeed that.

Each observation period, scheduled for a particular and different time each day, was of 60 minutes duration (in contrast to the 30 minute periods in the field). Data were compiled by noting the position of the infant at half-minute intervals and recording whether and with whom it was in contact, distance from the mother, and manner of behavior, particularly whether grooming or being groomed. These data themselves were grouped into 30 minute data periods that facilitated the various comparisons.

The animals were introduced into their chambers at least several weeks prior to the birth of an infant. Formal recording of data commenced upon the discovery of a newly-born infant, usually within hours, always within 24 hours after birth. Upon attaining 150 days of age, the infant was subjected to the First Separation, which involved removal of its mother to separate quarters for a 7 day period. At 200 days, there was a Second Separation also of 7 days' duration. Observations were made just before and after the removal and return of the mother. At 215 days the observations were terminated.

At 150 days, the infants could sustain themselves with solid food, though nursing was still occasionally observed. By 200 days, weaning was complete.

Results

Infants Alone with Mothers

The two cases of infants reared alone with their mothers were Sal, with her female infant, Ursula, born 3 April, 1969 and Calypso, with her male infant, Creon, born 20 March, 1970. Sal and Ursula were separated only once, at 200 days.

Figs. 3—6 reveal the changes in the maximum distances between mother and infant at various ages and the proportion of the observation time the infant was not in physical contact with its mother.

The trends are strikingly similar: both infants remained close to their mothers for the first 80 days. They never ventured even a meter from their mothers until they were over 50 days old, and both were about 110 days before they began maximally utilizing the space at their disposal. The proportion of time spent in physical contact with their mothers also did not show a significant decline until after 100 days age (by when about 1/3 of their "active" time was spent away from mother). No striking changes are indicated following the separations.

Tables 3 and 4 indicate how grooming behavior was influenced by the separations. Each table depicts the percentage of the 30 second intervals at

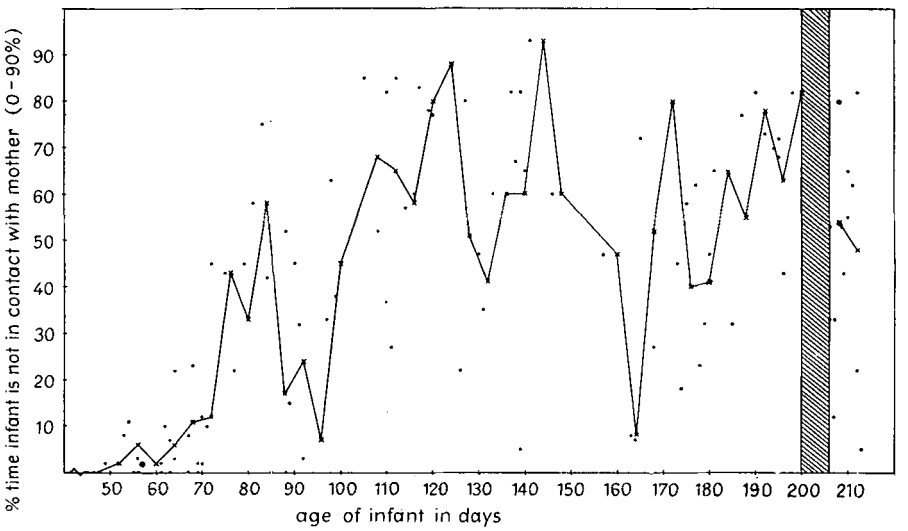


Fig. 3

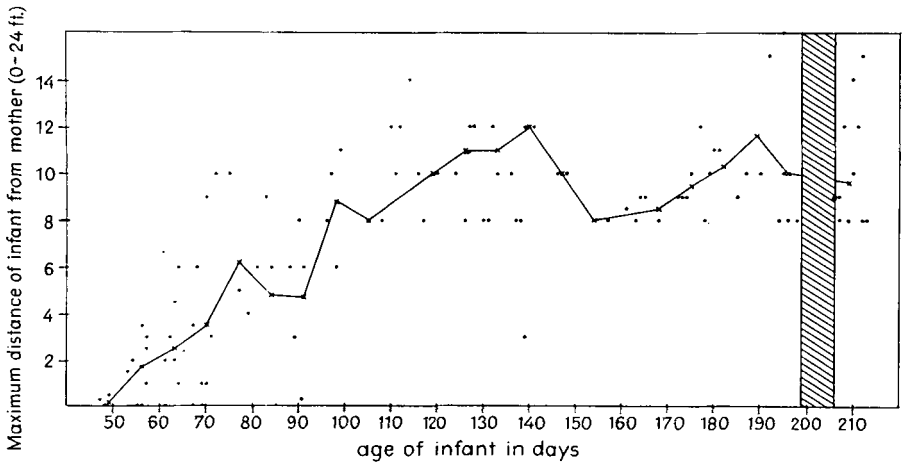


Fig. 4

the start of which grooming occurred during the week preceeding and following separation. Following the first separation, infant Creon markedly increased his grooming attentions to his mother, though the latter did not reciprocate. At the older age of the 2nd separation, the increase was less marked, though it occurred in infant Ursula's case, too. The impression given the observer

Table 3: Grooming Frequencies (% of 30 second observations) Sal and Ursula

	Female Grooming Infant	Infant Grooming Female	Mutual Grooming	Total	Infant Self-Grooming	Female Self-Grooming
Pre-separation week	1.3	4.4	7.3	13.0	9.8	1.8
Post-separation week	2.0	5.2	5.2	12.4	4.2	1.3

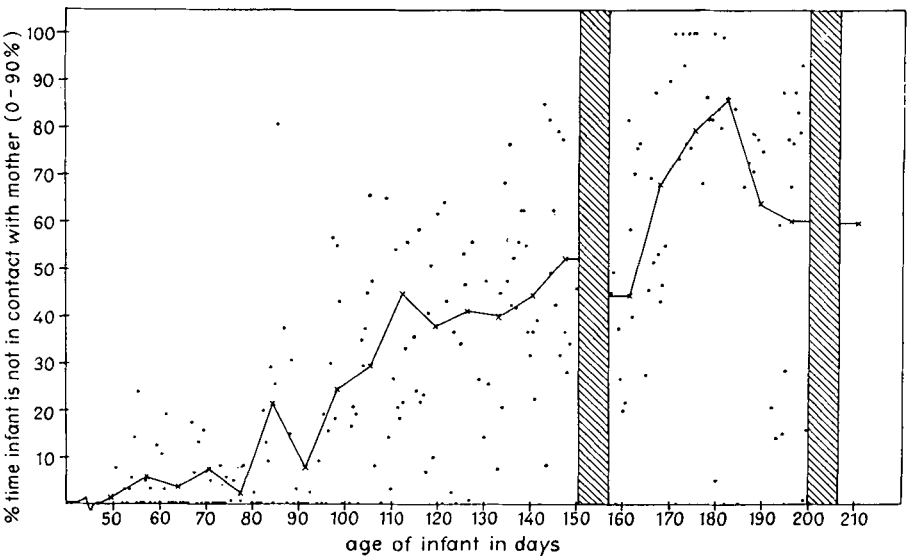


Fig. 5

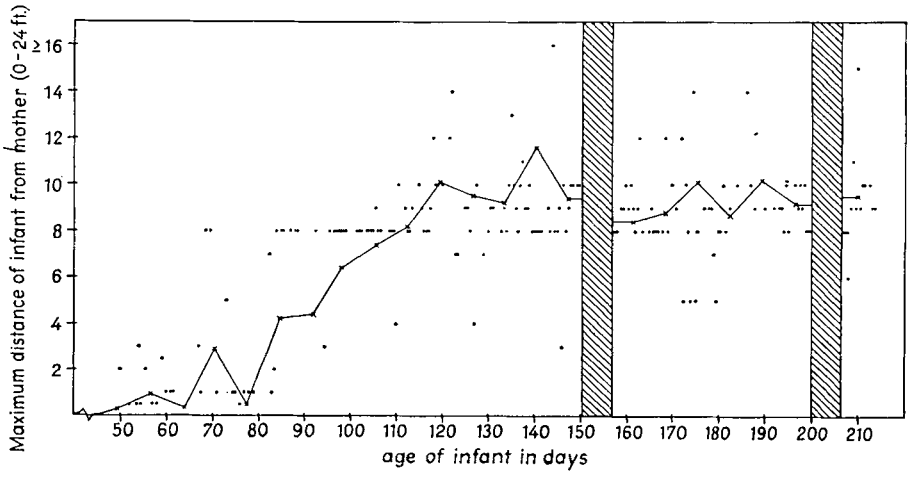


Fig. 6

Table 4: Grooming Frequencies (% of 30 second observations) Calypso and Creon

	Female Grooming Infant	Infant Grooming Female	Mutual Grooming	Total	Infant Self-Grooming	Female Self-Grooming
First separation						
Pre-separation week	2.3	0.8	3.3	6.4	6.9	9.2
Post-separation week	0.3	6.0	2.4	8.7	10.2	5.2
Second separation						
Pre-separation week	3.1	2.4	3.0	8.5	7.0	5.9
Post-separation week	3.8	4.1	7.5	15.4	11.5	5.5

was that the mothers were somewhat less than delighted at being returned to their infants while the latter were overjoyed. This response was probably dependent on it being specifically the mother and not some other animal as will be seen below, where the same pattern was obtained even with an affectionate male present. During the separation, incidentally, the mothers were kept solitarily and were equally deprived of grooming opportunities as their infants.

A separation at 150 days age was only effected in infant Creon's case. The observers' impressions at that time were of a frantic, hyperactive and highly vocal infant. The bouts of activity—wild, aimless chases about the room—were much briefer then usual, and interspersed with periods of stupor. Feeding habits deteriorated—the infant are little and with apparent disinterest. The picture was much the same during the 200 day separations, the first such separation for Ursula, the second for Creon. Both infants alternated between what appeared to be catatonic states with frenzied bouts of activity. This activity had a compulsive, stereotyped cast, and lacked the spontaneity of play.

On the return of the mothers, their infants invariably took the initiative in establishing contact. The mothers seemed nervous, examined the room minutely, and spent much time anal-marking limbs and platforms before attending to their infants. It was a full hour after her return before Calypso engaged in grooming with her infant, for example (1st Separation). A general restlessness on the mother's part persisted for a few days. By the end of the week following their reunion, however, no differences were noted in the daily patterns of activity and relationship.

Infants with Both Parents

The two cases of infants reared with both their parents were (1) Lisa with her female infant, Viola, born 14 April, 1970, and the male, Pop and (2) Atalanta with her female infant, born 12 May, 1969, and the male

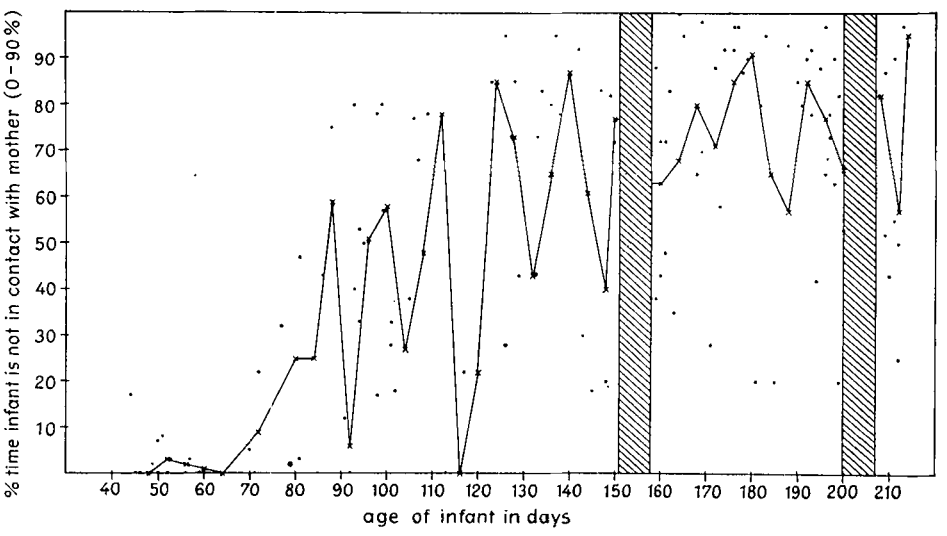


Fig. 7

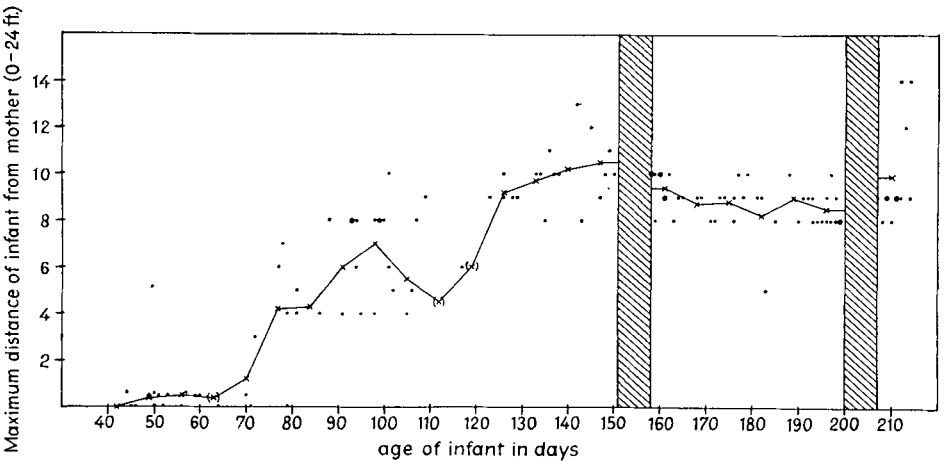


Fig. 8

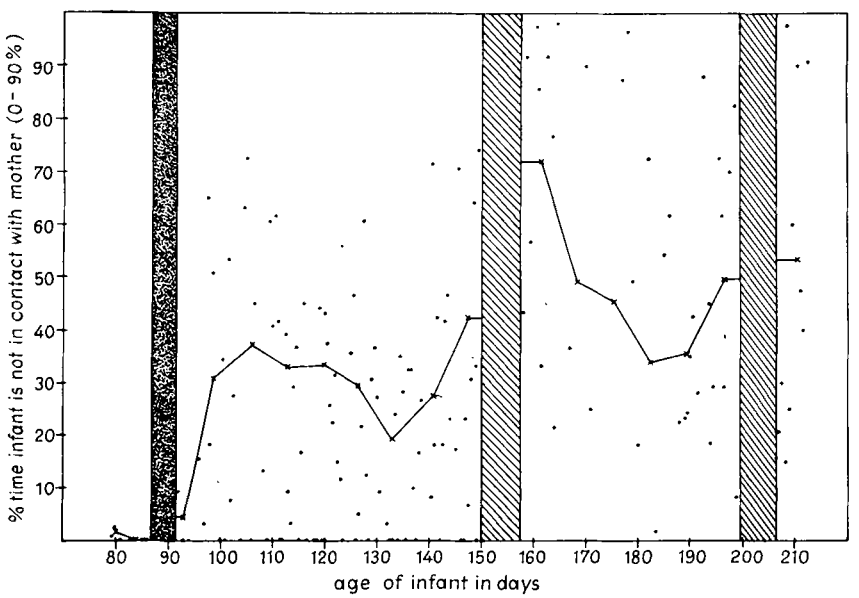


Fig. 9

Sforza. Lisa had actually given birth to twins, but one of these, the male Sebastian, was removed after 70 days due to illness. Thereafter, his mother declined to reaccept him and he was removed altogether. Figs. 7—10 reveal the changes in the maximum distances between mother and infant at various ages and the proportion of the observation times the infant was not in contact with its mother.

The early data from mother Lisa and infant Viola are difficult to interpret due to the dislocation occasioned by infant Sebastian's illness and removal, and a brief illness and separation of infant Viola. Curiously, these 3—4 day separations led the mother to subsequently reject and even attack her infants. The rejection of Viola gradually diminished, and she was finally

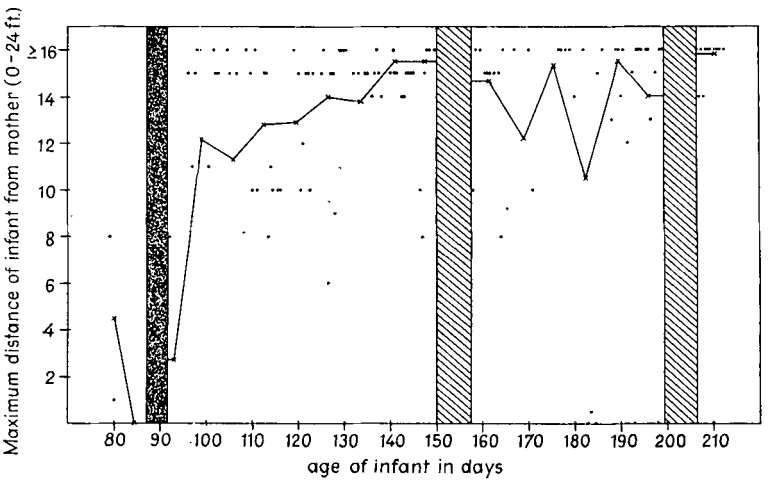


Fig. 10

re-accepted. Sebastian, however, was never re-accepted. The figures provide details of the temporal sequence. The early data are not plotted here since the infants could not be individually distinguished at the distance from which the observers had to work. In all events, the observations of both these family groups did show a similar pattern to that noted previously. For the first 70—80 days the infants remained very close to their mother, the first significant separation (one meter) not occurring until after the 70th day. However, the very first occasion of any kind of separation occurred almost 2 weeks earlier, on day 30, than among the "mothers only" group. Again, maximum utilization of the full area of the room, irrespective of the mother's locus, did not occur until after approximately 110 days. A sharp decline in the proportion of time spent in contact with the mother was noted by day 80 in Aphrodite's case, though not until day 100 in Viola's. The latter datum is of questionable significance due to the intervening illnesses. It does appear possible, however, that the presence of another animal does increase slightly the frequency of the infant's earliest departures from its mother. Again, the separations occasioned no striking changes in the relation to the mother. However, during the mother's absence there was a greatly increased amount of contact with the ♂. This is also illustrated by the graphs.

Tables 5 and 6 indicate how grooming behavior was influenced by the separations. In the case of infant Viola's first separation from her mother, the frequency of grooming of and with the ♂ increased during her mother's absence, falling upon mother Lisa's return. Upon Lisa's return, however, there was no corresponding increase in her grooming encounters with the infant, rather a slight decrease. A different pattern was seen during the second separation. Again, Viola greatly increased grooming contacts with the ♂ while her mother was absent, but on the mother's return Viola's grooming of her and mutual grooming between mother and infant rose from their pre-separation level. Mother Lisa's grooming of infant Viola, however, remained lower than prior to the separation.

With regard to Atalanta, Aphrodite, and Sforza, here, too, the first removal of the mother, Atalanta, occasioned a significant rise in the grooming between the male Sforza and the infant Aphrodite, which subsided on

Table 5: Grooming Frequencies (% of 30 second observations) Lisa, Pop and Viola

	Female Grooming Infant	Infant Grooming Female	Mutual Grooming	Total	Male Grooming Infant	Infant Grooming Male	Mutual Grooming	Total
First Separation								
Pre-separation week	3.4	1.2	1.6	6.2	0.3	2.2	0.7	3.2
Separation week					0.3	2.8	1.4	4.5
Post-separation week	2.6	0.4	0.8	3.8	0.3	1.0	1.1	2.4
Second Separation								
Pre-separation week	2.3	0.8	0.8	3.9	0.6	0.5	0.1	1.2
Separation week					0.3	4.8	3.3	8.4
Post-separation week	1.3	4.0	4.5	9.8	0.2	1.0	0.3	1.5

Table 6: Groomings Frequencies (% of 30 second observations) Atalanta, Sforza, and Aphrodite

	Female Grooming Infant	Infant Grooming Female	Mutual Grooming	Total	Male Grooming Infant	Infant Grooming Male	Mutual Grooming	Total
First Separation								
Pre-separation	1.4	0.7	0.0	2.1	0.5	0.2	1.0	1.7
Separation week					1.7	0.1	2.4	4.2
Post-separation week	3.0	0.9	2.4	6.2	0.4	0.3	0.8	1.5
Second Separation								
Pre-separation week	2.7	0.5	0.8	3.9	0.0	0.9	0.0	0.9
Separation week					0.6	2.2	3.3	6.1
Post-separation week	2.8	2.0	2.1	6.9	0.1	1.2	0.1	1.4

Atalanta's return. Her return did bring a rise in her grooming interactions with the infant, in contrast to what occurred in the other group. During the second separation there was again a transitory rise in grooming contacts with the ♂ (from 8 to 46 to 12), and also a rise in female—infant contacts on the mother's return (from 35 to 62). Once again, this represents the increased initiative taken by the infant. Occasions where the mother groomed normally ensued. Note particularly Fig. 11 with regard to this point.

A general picture of the behavior of the animals is best provided by the observers' weekly summaries, prepared from the daily notes and data sheets:

"Week 1. Mother Atalanta, father Sforza, infant Aphrodite at 143—150 days:

The infant moves independently of ♀ m, although tends to watch ♀ m, and will go to her if frightened by a sudden noise, etc. Infant initiated and breaks all contacts with ♂, and initiates most contacts with ♀ m. I—♀ m contacts are broken by either with about equal frequency.

The customary, almost invariable, sleeping position is: ♀ on top branch of tree, with infant in lap, nursing. ♀ faces the trunk and holds on with all of "hand". ♂ curls on top of the trunk, thus only a few inches from ♀ and I, but not in contact.

One instance seen of mutual boxing and play-fighting between ♀ and I, initiated by ♀ pulling infant's tail (unusual). Several "hit and run" incidents where infant → ♂ (and less often ♀), pulls tail, or hits, and dashes off—♂ usually ignores, sometimes snaps or hits back. Spats between ♀ and ♂ are common, though less frequent than when infant was younger. ♀ usually initiates, ♂ may retaliate, but finally yields.

♀ is first at food tray, infant follows her almost immediately, ♂ waits until they have gone to the tree with food before he goes to food tray. Infant often tries to take ♀ m's food, even when plenty remains in food tray. ♀ is generally tolerant, often sharing food with infant for a minute or so then relinquishing it. Sometimes ♀ snaps at infant but usually ends up letting infant have food and going back herself for more. Infant does not seem to try to take food from ♂.

Week 2. Aphrodite at 150—157 days.

♀m caught and removed at 9:30 a.m., October 10. While men were trying to catch ♀m, infant tried to cling to her, leaping off only when nets came really close. Immediately after ♀ was caught, infant leapt to ♂ and clung to him in the cross-ventral position for about 30 seconds, despite his frantic efforts to dislodge her. During the hour immediately following the removal of the ♀m, both infant and ♂ appeared to be quite agitated. There were very frequent movements together and apart with the ♂ taking initiative as often as infant; both were swinging their tails constantly, infant was giving calls—a rasping "quock-quock" at frequent intervals. (Infant calling heard as late in the week as 10/15.) In the first half hour, infant was cuddled on the ♂'s open lap (which was never seen in presence of ♀m) for 10 half-minute recordings—with much grooming of infant by ♂ and mutual grooming initiated by ♂.

Observed ♂ rubbing hands rapidly over branch-ends on 10/10, 10/11, 10/14.

After the first day (10/10) almost all contact was initiated by infant, and contacts were broken most often by infant but sometimes by ♂. Infant as active or more so than in preceding week. ♂ less sedentary than usual.

In absence of ♀m, I and ♂ were never observed to sleep on the previously habitual branch. The ♂ chose, instead, the next lower branch of the tree, were the baby would join him. (Had not observed any animal to sleep on this branch before.) Getting settled usually took a long time, (ten minutes on 10/14, 20 minutes on 10/15) with the pattern as follows: ♂ curls up on the branch, holding trunk. Infant comes, perhaps settles behind ♂ briefly then either tries to crawl in his lap or sits on top of his shoulders until uncurls or makes an open lap. ♂ may hold I in his lap and groom (observed on 10/11 for 1.5 minutes. on 10/13 for 3.5 minutes; on 10/15 for 2 minutes) but eventually (if not immediately) stands up over infant, breaking contact. Infant then moves to the next branch, ♂ curls up again in original position, I → ♂ and tries again to get in his lap. Infrequently, ♂ showed annoyance and snapped when infant trying to get in lap. Eventually infant settles behind ♂, embracing him and pillowing her head on his back. Slept so on 10/11, 10/13, 10/14, 10/15. On 10/12 they were observed for the first time sleeping on platform side by side facing room, infant facing wall, infant's arm over ♂'s back. This sleep pattern was also seen on 10/16 and 10/17.

Play-chasing, etc.—not observed during this week. ♂ snaps or hits at infant on occasion when infant tries to cling or climb into ♂'s lap, but no "playful" hitting or pulling of ♂'s tail and dashing off seen on part of infant.

Infant goes food tray immediately and ♂ holds back until people have left the wing. Then they visit food tray together and alternately, carrying food back to the tree to eat. On 10/14 infant was observed to try to take ♂'s food. They exchanged slaps and infant retreated without the food. Aside from this one observation, they were seen to feed near each other but without interaction.

Week 3. Aphrodite at 157—164 days.

When ♀m returned to room, there was a brief contact between infant and ♀m, then ♀m moved around the room sniffing and marking on branches and platforms. Infant followed ♀m, occasionally visiting ♂ very briefly. ♂ marked on infant once. ♀ approached ♂ several times, usually with head averted, seems very conciliatory, but ♂ avoided contact. ♂'s tail was swinging constantly.

In the second half hour after ♀ returned, ♀ made several more approaches to ♂. When infant was near ♂, ♂ left very quickly at approach of ♀ (who did not appear threatening). If infant not near, he stayed but longer, but still avoided contact with ♀. ♀m stopped sniffing and marking for one long mutual grooming session with infant on the original "resting branch" near top of tree.

Patterns of making and breaking contact were identical with Week 1. However, infant was seen to seek the ♂ lap on 10/19, and was accepted for 15 seconds before the ♂ moved. Infant is perhaps less independent of ♀ than in Week 1 in moving about the room, especially in early part of Week 3.

The most frequently seen pattern is identical with that of Week 1: ♂ on treetop, ♀ on highest branch of L. tree with I in lap and nursing. (Seen 10/18, 10/20, 10/22.) However, several other patterns were observed:

a) Same position as above, but infant behind ♀m (pick-a-back) (10/17).

b) ♀m with infant beside her, curled on platform 1, ♂ on branch of tree were he and infant habitually rested in Week 2 (10/17).

c) All together on platform 1—♀ laying over ♂'s back—infant lap ♀, and later pick-a-back ♀—all 3 in contact with each other (10/19).

d) ♀m and infant on platform 3, I in ♀'s lap, ♂ on L. treetop (10/21).

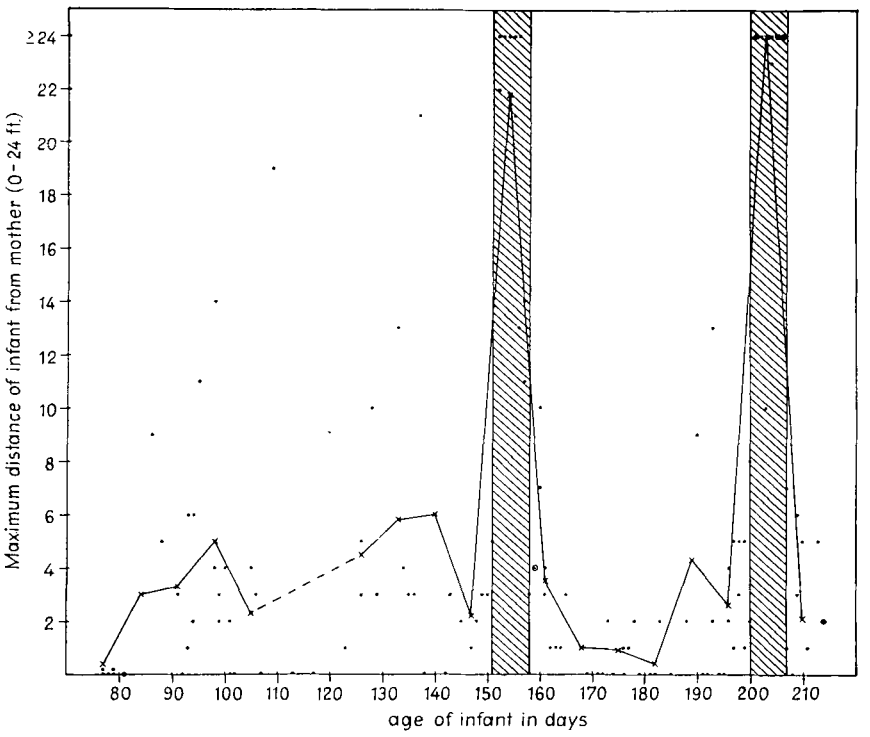


Fig. 11

Infant was observed to make a playful grab at ♂'s tail on 10/20—less frequent than Week 1. ♀m also is "nicer" to infant than in Week 1; there are fewer spats. On 10/23, ♂ approached resting ♀ with infant settled and nursing, and infant slapped ♂.

Feeding patterns identical with Week 1."

Table 6 provides additional support for the impression that no striking changes in the amount of contact between mother and infant followed their reunion. In the mother's absence the infant was by itself no more than previously (169/360 vs. 310/720—the 90% confidence intervals of these proportions overlap almost completely), though it greatly increased its contact with the father. In sum, the male merely replaced the female as a contact partner—there was no increase in total contact-time. On the mother's return, normally ensued. Note particularly Fig. 11 with regard to this point.

Infants with Extended Families

To date, detailed observations on the larger groups have been completed with but a single assemblage, mother Lisa, her infant, Corelli, born 6 February 1969, the ♂, Pop, and three adult ♀♀, Rufina, Clarissa, and Nausicaa. Further, after the first $2^1/_2$ months,

Table 7: Grooming Frequencies (Total Numbers of Separate Occasions During 7 Daily

	Female Grooming Infant	Infant Grooming Female	Mutual Grooming	Total
Pre-separation week	6	5	2	13
Separation week	-	-	-	-
Post-separation	1	8	14	23

this group was no longer housed within the standard observation chamber, but due to various practical problems had to be moved outdoors to a chain-link-fenced enclosure nestled within three sides of the Primate Facility (Fig. 1). The area of the enclosure is approximately 150 m². The floor is grass, and narrow boards provide aerial walkways between the vertical posts, set at approximately 3 meter intervals to support the overhead wire. Shelves with wind-protecting sides and infra-red heaters provide cover and warmth. Observations were made from behind the Facility's large glass windows which form part of one wall of the enclosure. These are of ordinary glass, so the observers were visible to the animals. However, the windows are along the service corridor linking the kitchen to the animal rooms. A steady stream of people is constantly in the animals' field of view, so rather complete habituation to their presence may reasonably be assumed to occur.

The difference in the manner in which the animals were housed was further exacerbated by differences in observational and recording procedures, this group having been studied prior to our having settled upon one standard method. Thus, all of the desired comparisons with the other groups cannot yet be made. A number of facts can nonetheless be usefully gleaned from the observations.

The infant was 46 days old when first observed off her mother. She crawled onto the ♂'s back at a distance of 1/3 meter from her mother. Separations became more frequent thereafter, and by 55 days these excursions had carried the infant up to 2 meters away. A steep increase in separation frequency began from this day. On day 74, the group was moved outdoors, as noted above, and for 3 weeks thereafter the infant remained close to her mother, then, rather abruptly, again separating herself as she had before.

A single separation of mother Lisa was conducted for one week at 150 days. The relative grooming frequencies are given in Table 7. These data are not directly comparable to those derived from the other animals due to differences in sampling methods. In the absence of her mother, however, the infant does markedly redirect her grooming activities to the ♂, rather than to the other ♀♀. Their grooming frequencies seem to change but gradually from week to week (and are quite similar to the frequency of the adult ♀ grooming frequencies *inter se*). As with the other groups, on the mother's return it is the infant that initiates most encounters, in this instance raising the total number of grooming interactions from 13 to 23. This infant did not correspondingly reduce her interactions with the ♂ on her mother's return, however. The observer's impression was that mother Lisa and father Pop had a joyous reunion, and that Lisa had little enthusiasm for her infant.

Table 8 summarizes, for all groups, the patterns of grooming response that are evident. The total amount of grooming does apparently rise following reunion. At least there was an increase in 6 of the 7 weeks of observation if we compare the pre- and post-separation periods. The frequently of the infants's grooming of its mother also increased in the same 6 of 7 cases, though, within each cell, this increase was significant in only 5 instances (Chi-square with $p = 0.95$). The impression of the observers that the separation diminished the ♀'s interest in her infant was not entirely supported by the grooming data. At least with regard to the frequency of the ♀'s grooming of the infant, there was a reduction as often as there was an increase in the post-separation week (4 instances of reduction, 3 of increase, and in only 1 case was the change significant, Chi square with $p=.95$). The amount

30 Min. Periods) Lisa, Corelli, Pop, and others

Male Grooming Infant	Infant Grooming Male	Mutual Grooming	Total	All Grooming With Others
2	4	2	8	12
12	8	6	26	7
2	10	8	20	9

Table 8: Summary of Grooming Data

	Sal, Ursula	Calypso, Creon	Lisa, Viola, Pop	Atalanta, Aphrodite, Sforza
a. Does the total amount of grooming between mother and infant rise after the separation?	+	first separation: +; second separation: +	−; +	+; +
b. Does the mother groom her infant more before the separation than after?	−	+; −	+; +	−; −
c. Does the infant groom its mother more after the separation than before?	+	+; +	−; +	+; +
d. Does the ratio of infant-grooming-mother / mother-grooming-infant increase after the separation?	−	+; +	−; +	−; +

of mutual grooming increased significantly in 5 of the 7 cells. This increase could, of course, be interpreted as indicating heightened interest in the infant by the ♀. However, since the frequency of infants grooming their mothers increased in addition to the increase in the mutual grooming episodes, it seems reasonable to conclude that the reunion indeed affect the infant more markedly than its mother.

Spacing as a Function of Substrate

By varying the number of animals in a group one inevitably also varies the space available to any one of them. Perhaps even more importantly, one alters the character of the available space. Wherever any degree of dominance or attraction exists between two or more animals, the area surrounding them must be viewed as charged, analogous to the surround of atomic nuclei. The shape of the space available to other animals will then depend on their valence and will be altered as they move about. Complexities of the substrate, partitions, overhanging ledges, holes, may function as insulators, distorting the fields of charge and permitting distributions that would otherwise be impossible. Hence, the experiments on the infant separation trauma and its amelioration require an assessment of the role of the substrate on social spacing.

The relation between space and social organization has been frequently studied with regard to man. Recent volumes by SOMMERS (1969) and ESSER (1971) provide especially keen discussions of the complexities of the issue. It is unlikely that the problem is much simpler in animals other than man, though surely its manifestations differ. A survey of the situation in primates, for example, will be found in CROOK (1970).

In the Gilboa Mountains of Israel, herds of gazelles (*Gazella gazella*) graze on slopes of the broad wadis. At their eastern end, near Bet Shean, visibility extends for several kilometers from top to bottom and along the wadis, for there are relatively few visual obstructions. The grass is short, brush or trees are generally confined to a narrow strip in the wadi bottom, and rocky outcrops are few. A few kilometers farther to the west, the wadis are narrower, their sides more precipitous, impediments to distant vision much more frequent. Preliminary observations of the gazelles suggest that these western herds generally move in linear fashion when alarmed. Widely spaced, and with distinctive "leadership" shown by the first and last animals. In contrast, the more easterly herds more often move along a broad front, with no particular animal obviously setting pace or direction. While our data on these gazelles are as yet incomplete, they do suggest how profoundly social structure may be influenced by substrate.

The forests of Madagascar also vary in their structure, from the broad, drooping-limbed Tamarinds (*Tamarindus indica*) to the vertically aligned, nearly limbless euphorb, *Didiera madagascariensis*. To what extent do differences in foliage density, limb structure,

and distribution influence behavior within any one species of lemur? (The question of interspecific differences poses a rather different problem, with which we shall not here be concerned.)

Methods

Four young *Lemur fulvus*, ♂♂ Cadmus (3/15/70) and Aeneas (4/12/70) and ♀♀ Clio (3/18/70) and Dido (3/21/70). These animals were caged with their mothers, then brought together (without their mothers) in an observation room as pictured in Fig. 2. Platforms were installed in 3 corners, a single 4 meter longer "2×4" (5 cm×10 cm) board from the center of the room to another corner and a wooden chair sat in the center. After 35 days habituation, a 30 day observation session began. At 2—4 predetermined intervals during each day, the positions and activities of each animal were noted. At the end of the 30 day session, the animals were briefly caged and their room "reforested", as indcated by Fig. 2. The reforestation involved the addition of 5 trees, and numerous horizontal limbs, including 3 that ran from tree to tree. The animals were released and observed over a second 30 day session. The total number of observation periods for each session was 113 and 116, respectively. These were reduced to 100 each by random drawing. An observation was scored "active" so long as one animal was moving about at the start of the observation. The animals showed a high degree of synchrony in their activity cycles.

Results

Fig. 12 indicates the number of observations in which the animals were active or resting, and, in the latter case, whether sitting alone or the size of the largest number in contact with one another.

The number of active (and resting) periods did not differ significantly between the two environments (53 and 54, for the "impoverished" and "enriched" rooms, respectively). There were small differences between rooms in the amount of clumping, however. All the animals were alone 13 times in the impoverished room, compared with 28 in the enriched room (confidence intervals of these proportions are separate with confidence coefficients between .90 and .95). Clusters of 2 animals were seen 72 (impoverished) versus 53 (enriched) times, confidence intervals of these proportions are separate with confidence coefficients between .90 and .95). Larger clusters were infre-

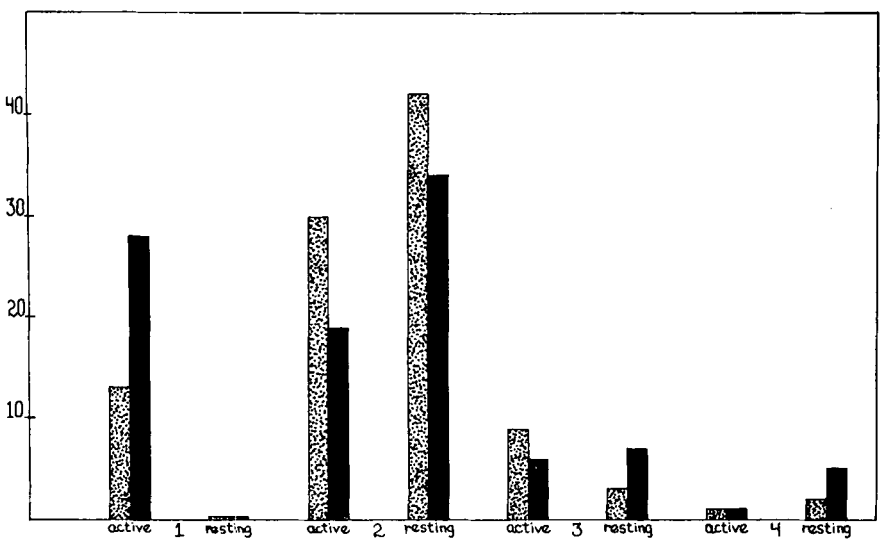


Fig. 12: Clumping in impoverished (stippled) and enriched (solid) environments

quent, and did not differ significantly, though during resting the enriched room seemed to promote more clumping (groups of 3 and 4, 12 *vs.* 5, difference too small to show significance, if any). The total number of animals involved in any size clump in all observations was 192 (impoverished) vs. 169 (enriched), which is not significant. Since the frequency of activity was unchanged between environments, differences in cluster frequency are not likely to be an artefact of the animals' speed or frequency of movement. The impoverished room did have fewer perches and rest areas, of course, though there was a sufficient number as to allow total separation (3 shelves, a chair, a feeding ledge). Separation did not occur as often: under impoverished conditions, the animals were more often paired (whether resting or active) than under enriched conditions. The large clumps, on the other hand, seemed to occur more often among resting, "enriched" animals. Hence, the effect of the artificial forest is not merely to provide a greater array of potential perches Were that the case, one would expect fewer, rather than an equal or greater number of the 3 and 4 animal clumps. Nor did the "forest" alter activity frequency, and thus, indirectly, the groupings. The observations themselves reveal no qualitative differences in the nature of the activities in which the animals engaged in the two environments. In a subtle, as yet undefined manner, the "forest" did cause small differences in the groupings to emerge: less dispersion of individuals in the impoverished room, and a predominance of pairs over other clumps of another size. (The ratios of the number of animals in trios or quartets to the number in pairs were 48/144 for the impoverished, 63/106 for the enriched groups; the confidence intervals of these proportions differ with coefficients between .90 and .95, $N = 100$.)

General Conclusions

Any number of studies involving primates have been reported in which the effects of separating mother and infant have been assessed, as well as the consequences of rearing young (with or without their mothers) in relatively impoverished environments (for instance, note HARLOW and HARLOW 1965; JENSEN, BOBBITT and GORDON 1968; SPENCER-BOOTH, and HINDE 1967; ROSENBLUM and KAUFMAN 1967). At this juncture, however, comparisons with these and other important studies would serve no useful purpose: that data presented here are themselves too fragmentary to even allow for generalizations regarding the Lemuriformes, let alone comparisons with the monkeys or apes. Hence, we shall have to remain content for yet another period with a modest description of the development of mother-young relations in captive lemurs, as inferred from the observations reported here. The answers to the larger questions towards which these observations are directed are not yet in hand.

In *Lemur fulvus,* it appears the infant is greatly disturbed by a week-long separation from its mother only if no other animal is available. Of course, a separation at an earlier age, prior to weaning, might well be more traumatic. The empirical problem in instituting earlier separations is the confounding of nutritional deficiencies or hand-feeding by humans. It is also questionable as to whether infants less than 150 days of age are, in nature, ever subject to t e m p o r a r y separations. By the time they attain an age of 150 days, on the other hand, their mothers will be coming into estrus. The excitement of the breeding period is rather more likely to lead to short separations between mothers and their infants.

Because of the relatively great duration of the period in which the infant is in contact exclusively with its mother (in contrast to *Lemur catta,* for instance), no great effect of rearing mother-infant pairs in isolation from other animals was expected. Our results suggest, contrary to expectation, that having a male present increased slightly the rate at which the infant became independent. In the presence of yet more animals, an even greater acceleration appears to occur. JENSEN (1968) had noted that the mother-young relation was intensified and development retarded by rearing in stimulus-poor environments (in macaques). On the other hand, SPENCER-BOOTH, HINDE, and BRUCE (1965) found the macaque infants ranged more from their mothers when under (comparable?) conditions of isolation.

These differences obviously can have a variety of explanations. Additional data, particularly from *L. catta* and *L. variegatus,* will be needed before speculation becomes profitable. Fortunately neither confinement nor the impoverished character of the physical environment inhabited by our captive lemurs appear to drastically alter the relations between a mother and her infant.

Acknowledgments

This work was done at the Primate Facility of Duke University's Field Station for Animal Behavior Studies, supported by NSF GB4000 and NIH FR00388 and HD02319, as well as a Research Scientist Award. I also record my appreciation to Diane CHEPKO, Jenny GREGG, Linda HOEBET, Martha KLOPFER, and Janice PULLIAM for their many hours of observation and compilation. The good health of our colony is largely due to the ministrations of Dr. Jan BERGERON and his staff. The observations of gazelles were made possible by the thoughtfulness and kindness of Drs. H. MENDELSSOHN and Yoram YOM-TOV, of the University of Tel Aviv.

We must here record our indebtedness to Monsieur Henri de HEAULME et fils, not merely for the use of their facilities at Berenty, but also for their efforts to preserve the habitat and the lives of the lemurs. We acknowledge, too, the efforts of the Comite de Scientifique et Recherche de Repoblika Malagasy.

The light amplification device for nocturnal observations was kindly loaned by Dr. Arthur J. EMERY, Scientific Analysis Branch, Arlington, Virginia.

Summary

Observations of *Lemur catta* mother-infant pairs in a forest preserve in Madagascar provided assurance that the conditions of captivity in the Duke University colony were not seriously distorting the behavioral relations between mothers and their infants. Intensive observations of captive, individual mother-infant pairs of *L. fulvus* suggested that the frequency and distance of the infant's excursions from its mother were increased with the presence of other lemurs. The temporary absence of the mother upon the infant's attaining an age of 150 days was readily tolerated if another familiar animal was present, but otherwise proved traumatic. However, no subsequent effect of a separation of 1 week's duration was noted. On reunion, the infants seemed more eager to reestablish contact than their mothers.

Comparisons with other primates, and speculations respecting the relation of infant rearing behavior to the ecology and dispersal habits of the different species of lemur, will require additional data, particularly on *L. catta* and *L. variegatus.*

Zusammenfassung

Freilandbeobachtungen in Madagaskar zeigten, daß die Haltungsbedingungen für Mutter-Kind-Paare von *Lemur catta* an der Duke Universität das

Verhalten der Tiere nicht wesentlich verzerren. Beobachtungen an *Lemur fulvus* ergaben, daß Junge öfter und weiter von ihrer Mutter weggingen, wenn andere Lemuren anwesend waren. 150 Tage alte Junge überstanden eine zeitweise Abwesenheit der Mutter ohne Trauma, wenn ein anderes vertrautes Tier da war. Doch gab es keine Nachwirkungen einer 1-wöchigen Trennung. Nach der Trennung schienen die Jungen mehr den Kontakt mit der Mutter anzustreben als umgekehrt.

Literature cited

BUETTNER-JANUSCH, J. (1966): The Origins of Man. J. Wiley & Sons • BUETTNER-JANUSCH, J. MS: The Lemurs • CROOK, J. H. (1970): Social Behavior in Birds and Mammals. Academic Press • ESSER, H. H. (1971): Behavior and Environment. Plenum • HANSEN, E. W. (1966): The development of maternal and infant behavior in the Rhesus monkey. Behaviour **27**, 107—149 • HARLOW, H. F., and M. K. HARLOW (1965): The affectional systems. In: Behavior of Non-human Primates (A. M. SCHNIER, H. F. HARLOW, & F. STOLLUITZ eds.), 287–334, Academic Press • JENSEN, G. D., R. A. BOBBITT, and B. N. GORDON (1968): Effects of environment on the relation between mother and infant pig-tailed monkeys. J. Comp. Physiol. Psych. **66**, 259—263 • JOLLY, A. (1966): Lemur Behavior. University of Chicago Press • KLOPFER, P. H., and M. S. KLOPFER (1970): Patterns of maternal care in lemurs I: Normative description. Z. Tierpsychol. **27**, 984—996 • KLOPFER, P. H., and A. JOLLY (1970): The stability of territorial boundaries in a lemur troop. Folia. Primat. **12**, 199—208 • PETTER, J. J. (1965): The lemurs of Madagascar. In: Primate Behavior (I. de VORE ed.), 292—321. Holt, Rinehart & Winston • ROSENBLUM, L. A., and I. C. KAUFMAN (1967): Laboratory observations of early mother-infant relations in Pig-tail and Bonnet Macaques. In: Social Communication in Primates (S. A. ALTMANN ed.), 33—42. University of Chicago Press • SOMMERS, R. (1969): Personal Space. Prentice Hall • SPENCER-BOOTH, Y., and R. A. HINDE (1967): The effects of separating rhesus monkey infants from their mothers for six days. J. Child Psychol. Psychiatr. **7**, 179—197.

Author's address: Peter H. Klopfer, Zoology Department, Duke University, Durham, North Carolina 27706, U.S.A.